DP-DBBA Evaporation

Phase 03 | Operator explanation and control logic

Release	Build	Reviewed	Executable
0.9.41	2026.09.04-r20	2026-09-04	Phase 03

Safety boundary: this guide describes the software sequence only. Follow the current laboratory SOP, confirm the physical chamber and instruments, and remain supervised. Software limits do not replace hardware interlocks or the emergency procedure.

Purpose

Evaporate DP-DBBA at the sample-equivalent target derived from the confirmed Phase 01 thickness ratio, with the external oven qualified before shutter opening.

Scope and ownership

The phase owns CK-1 control, sample/QMB tracking, external oven readiness, shutter gating, plots and safe shutdown. It is started only after the launcher has shown and confirmed a valid ratio.

Operator sequence

1	Review the latest saved Phase 01 ratio shown by the launcher. Select Yes only if it matches the physical run; otherwise enter the ratio manually.
2	Confirm the DP-DBBA sample-equivalent target and complete the physical chamber and instrument pre-flight.
3	Start the phase. The external oven PID is set to 200 deg C and must remain within +/-2 deg C for 60 continuous seconds.
4	Supervise CK-1 warm-up using the selected validated feedback mode. The default guide is 242 deg C and 0.40 angstrom/s.
5	Open the shutter only when the external oven readiness, CK-1 temperature, averaged CK-1 rate and ratio-dependent checks are satisfied.
6	Close the shutter when the target evaporation amount is reached or the dashboard prompts. The graph records the exact open/close interval and relative thickness traces.
7	Use Finish for the documented Phase 04 handoff, or Abort / Safe Stop if the process must stop immediately.

Packaged defaults

Parameter	Default	Meaning
Sample-equivalent target	623.13 / 94.39 angstrom	Amount multiplied by the confirmed Phase 01 ratio.
External oven target	200 deg C	Startup setpoint before evaporation readiness.
Oven readiness	+/-2 deg C for 60 s	Independent external-oven prerequisite.
CK-1 temperature guide	242 deg C	Process guide and temperature-mode target.
CK-1 rate guide	0.40 angstrom/s	Averaged rate gate and rate-mode target.
Automatic current cap	0.660 A	Highest editable automatic command.
Software hard stop	0.680 A	Fixed current stop.
Ratio source	Phase 01 or manual	Must be explicitly confirmed before start.

Shutter gate and measurement

The shutter is a process boundary, not a controller-tuning diagnostic. Phase 03 requires the external oven readiness condition and the minimal CK-1 process conditions before it can request shutter opening. The ratio-dependent target is calculated from the confirmed calibration ratio and saved with the run.

Live and saved plots show absolute thickness and shutter-relative thickness. The relative traces start at 0 angstrom at the open-shutter timestamp; raw measurements and summaries remain available for audit.

> If the ratio is uncertain, stop and resolve it before opening the shutter. A manually entered ratio is an operator decision and must be recorded with the run documentation.

Finish and abort behavior

Normal completion	Abort / Safe Stop
After shutter close and normal completion, Keysight current returns to 0.640 A and the output remains ON for the Phase 04 handoff. Final files are saved before the phase exits.	Abort commands Keysight current 0 A and output OFF immediately. The oven PID reset to 0 deg C is attempted afterward on a best-effort basis so it cannot delay the electrical shutdown.

Data written

A new folder is reserved below Data Samples/DP-DBBA Evaporation Data using the pattern DP-DBBA Evaporation <sample> data NN.

Important record	Use
Effective parameters	Confirmed ratio, target calculation, ramp/PID settings and fixed safety values.
Thickness summary	Absolute and shutter-relative CK-1/sample thickness and the final amount.
Plots and snapshots	Oven readiness, QMB signals, currents, rates, pressure and shutter markers.

Handoff

Do not start Phase 04 until the launcher reports that Phase 03 has exited and all serial ports have been released. Phase 04 expects the documented 0.640 A Keysight handoff state.